Web Development using PHP

Brief overview of web development and HTML

(Lecture 1)

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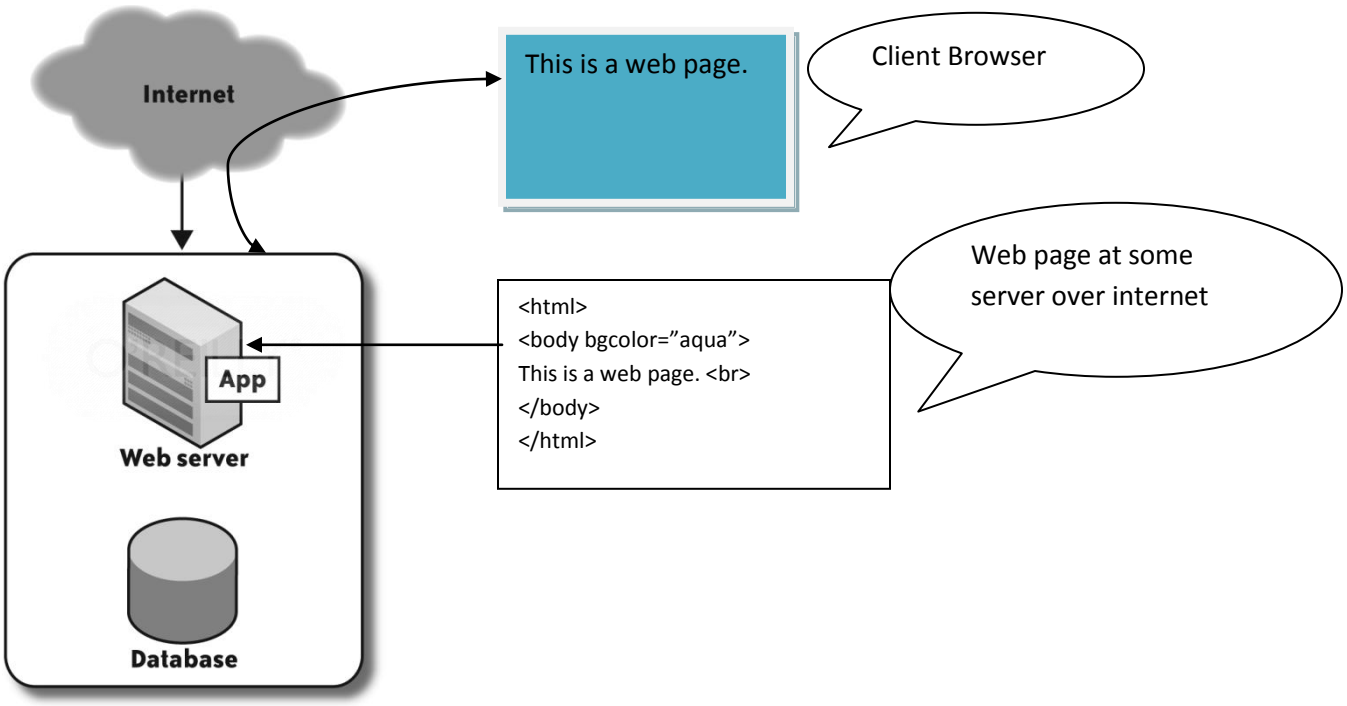
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1. Web Development Architecture

It typically comprises of a browser, HTML, any scripting language, web server and database server. In web development we use all of these to display some information (or build a page) to our target user over the internet. The general pattern of displaying information is as follows;

When a request for a page comes from the browser, the web server performs following steps:

1. Read the request from the browser.
2. Find the page on the server.
3. Send the page back across the internet to the client browser.
4. Client browser interprets HTML and create (display) the (information requested) as a web page.



1.1 Browser

The purpose of a web browser (like Internet Explorer or Firefox) is to read HTML documents and display them as web pages. The browser does not display the HTML tags, but uses the tags to interpret the content of the page.

1.2 Web Server

The primary function of a web server is to deliver web pages on the request to clients. This means delivery of HTML documents and any additional content that may be included by a document, such as images, style sheets etc.

2. HTML (Hyper text markup language)

HTML is a language for describing web pages.

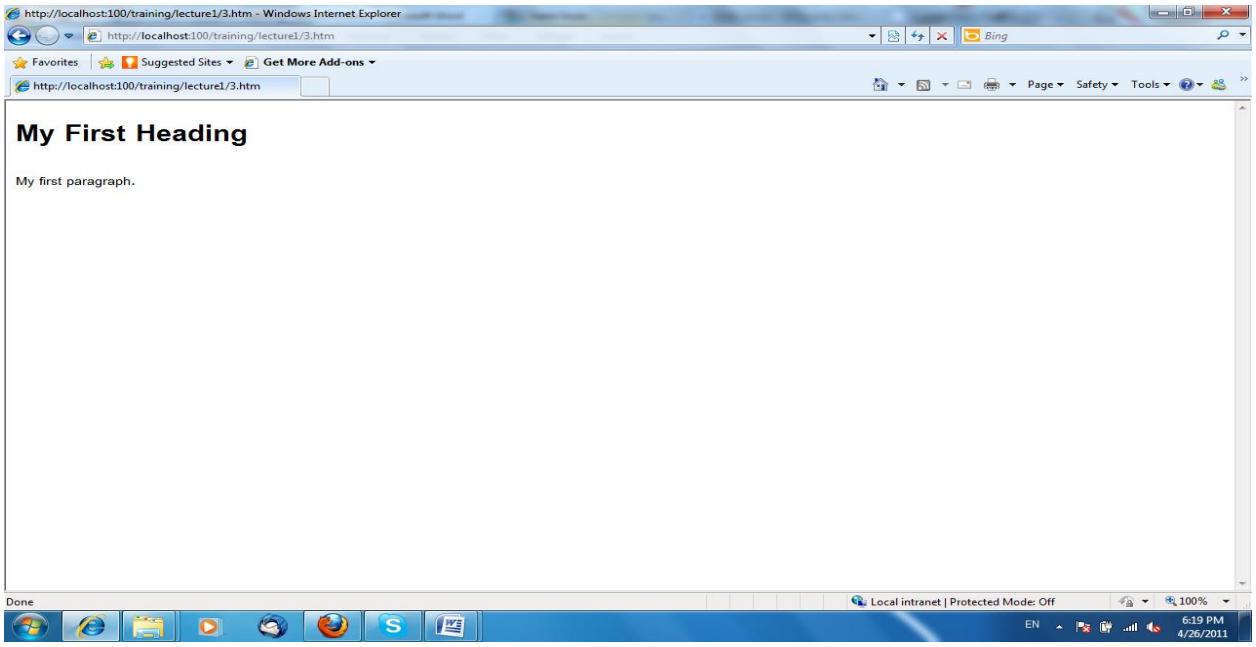
- HTML stands for **H**yper **T**ext **M**arkup **L**anguage
- HTML is not a programming language. It is a **markup language.**
- A **markup language** is a modern system for annotating a text in a way that is syntactically distinguishable from that text.
- HTML uses **markup tags** to describe web pages

```
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

Example 2.1 (Explained)

- The text between <html> and </html> describes the web page
- The text between <body> and </body> is the visible page content
- The text between <h1> and </h1> is displayed as a heading
- The text between and is displayed as a paragraph

Its output would be;



3. Styling HTML with CSS

CSS was introduced with HTML 4, to provide a common way to style HTML elements.

CSS styling can be added to HTML in the following ways

- in **separate style sheet files** (CSS files)
- in the **style element** in the HTML head section
- in the **style attribute** in single HTML elements.

Example 3.1 (HTML Style- Background Color)

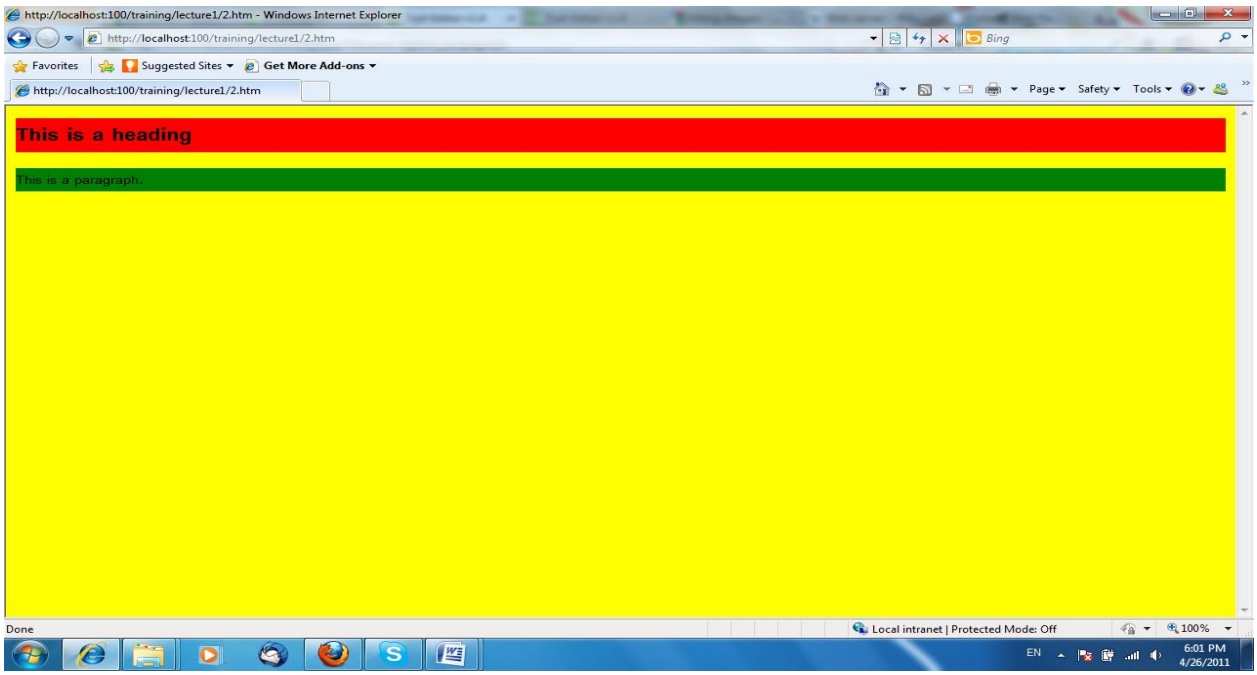
The background-color property defines the background color for an element:

```
<html>

<body style="background-color:yellow">
<h2 style="background-color:red">This is a heading</h2>
<p style="background-color:green">This is a paragraph.</p>
</body>

</html>
```

This outputs as follows;



Example 3.2 (HTML Style - Font, Color and Size)

The font-family, color, and font-size properties define the font, color, and size of the text in an element:

```
<html>

<body>
<h1 style="font-family:verdana">A heading</h1>
<p style="font-family:arial;color:red;font-size:20px;">A paragraph.</p>
</body>

</html>
```

[Try it yourself]

4. Active Pages

The web has now been moving towards active or dynamic websites, which allow users to be sent customized pages and which offer more dynamic browsing experience. These are built with a combination of languages and technologies and we can use any one of them.

We can split these technologies into two groups: client side and server side technologies. The former include;

1. javascript

2. Active X Controls

The server side technologies include;

1. CGI
2. ASP
3. JSP
4. PHP
5. Coldfusion

4.1 Scripting Language PHP

- PHP stands for PHP: **H**ypertext **P**reprocessor
- PHP is a server-side scripting language, like ASP
- PHP scripts are executed on the server
- PHP supports many databases (MySQL, Informix, Oracle, Sybase, Solid, PostgreSQL, Generic ODBC, etc.)
- PHP is an open source software
- PHP is free to download and use

PHP was developed in 1994 by Rasmus Lerdorf to track visitors to his online resumes and was released as Personal Home page tools the following year. This was rewritten and combined with an HTML Form Interpreterin 1995. This grew rapidly in popularity and by around the middle of 1997. PHP had ceased to be Rasmus Lerdorf's personal project and had become an important web technology. The parser was completely rewritten by Zeev Suraski and Andi Gutmans, and PHP3 was released in June 1998.

PHP works in a similar way to JSP and ASP: scripts sections are enclosed in <?php .. ?> tags and embedded within an HTML page. These scripts are executed on the server before the page is sent to the browser, so there is no issue of browser support for PHP page. The decisive factor is that it's free and open source.

4.2 Benefits of Server Side processing

1. Minimizes network traffic.
2. Make quicker downloading since the client browser only downloading HTML page.
3. Can provide data that does not reside at the client.
4. Provides improved security measure, since we can code things that can never be viewed from the browser.

4.3 What is a PHP File?

- PHP files can contain text, HTML tags and scripts
- PHP files are returned to the browser as plain HTML

- PHP files have a file extension of ".php", ".php3", or ".phtml"

4.4 Where to Start?

To get access to a web server with PHP support, you can:

- Install Apache (or IIS) on your own server, install PHP and ready to go.

5. Scripting Language Javascript

- JavaScript was designed to add interactivity to HTML pages
- JavaScript is a scripting language
- A scripting language is a lightweight programming language
- JavaScript is usually embedded directly into HTML pages
- JavaScript is an interpreted language (means that scripts execute without preliminary compilation)
- Everyone can use JavaScript without purchasing a license

5.1 Are Java and JavaScript the same?

NO!

Java and JavaScript are two completely different languages in both concept and design!

Java (developed by Sun Microsystems) is a powerful and much more complex programming language - in the same category as C and C++.

JavaScript was invented by Brendan Eich at Netscape (with Navigator 2.0), and has appeared in all browsers since 1996.

5.2 What can a JavaScript do?

- **JavaScript gives HTML designers a programming tool -** HTML authors are normally not programmers, but JavaScript is a scripting language with a very simple syntax! Almost anyone can put small "snippets" of code into their HTML pages
- **JavaScript can put dynamic text into an HTML page -** A JavaScript statement like this: document.write("<h1>" + name + "</h1>") can write a variable text into an HTML page
- **JavaScript can react to events -** A JavaScript can be set to execute when something happens, like when a page has finished loading or when a user clicks on an HTML element

- **JavaScript can read and write HTML elements -** A JavaScript can read and change the content of an HTML element
- **JavaScript can be used to validate data -** A JavaScript can be used to validate form data before it is submitted to a server. This saves the server from extra processing

6. Ajax

AJAX = Asynchronous JavaScript and XML.

AJAX is not a new programming language, but a new way to use existing standards.

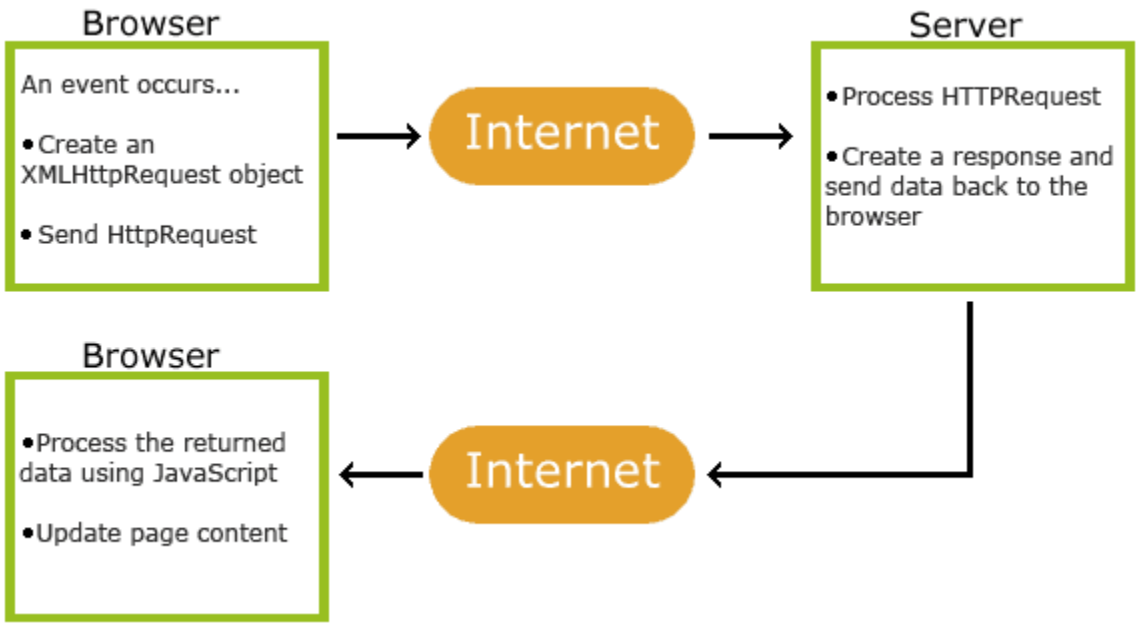
AJAX is the art of exchanging data with a server, and update parts of a web page - without reloading the whole page.

AJAX allows web pages to be updated asynchronously by exchanging small amounts of data with the server behind the scenes. This means that it is possible to update parts of a web page, without reloading the whole page.

Classic web pages, (which do not use AJAX) must reload the entire page if the content should change.

Examples of applications using AJAX: Google Maps, Gmail, Youtube, and Facebook tabs.

6.1 How Ajax works



AJAX is based on internet standards, and uses a combination of:

- XMLHttpRequest object (to exchange data asynchronously with a server)
- JavaScript/DOM (to display/interact with the information)
- CSS (to style the data)
- XML (often used as the format for transferring data)

7. Using Databases

Databases are used to store information and scripting languages provide ways to access this information manipulate and send to the client browser. We'll talk the PHP way and capability to interact with databases and manipulation in coming lectures. For now we'll see the most basic way which PHP used to connect with a database.

Before you can access data in a database, you must create a connection to the database.

In PHP, this is done with the [db type]_connect() function.

7.1 Syntax

[db type]_connect(servername, username, password);

Parameter	**Description**
servername	Specifies the server to connect to.
username	Specifies the username to log in with. Default value is the name of the user that owns the database server
password	Specifies the password to log in with.

Example 7.1

In the following example we store the connection in a variable ($con) for later use in the script. The "die" part will be executed if the connection fails:

```
<?php
$con = oci_connect("db user","db pass","oracle sid");
if (!$con)
 {
 die('Could not connect: ' . oci_error());
 }

// some code
?>
```

7.2 Closing a Connection

The connection will be closed automatically when the script ends. To close the connection before, use the oci_close() function. In the above example, we can close the connection as follows;

oci_close($con);

8. A simple web page program layout /structure

```
<html>
<style>
some style definitions
</style>

<script language='javascript'>
some javascript code
</script>

<body>
some html code here

<?php
// some code
?>
</body>
</html>
```

9. Building an HTML Page

Now we'll create various HTML on a web page. So please create a web page (blank page) with yourname.htm in your htdocs/training/lecture1/ directory.

Now Edit the file and Put the following code.

Example 9.1

```
<html>
<body>
</body>
</html>
```

run the page in the browser by typing http://localhost/training/lecture1/yourname.htm

See what happens or displayed there.

Now add following line between html and body tags.

Example 9.2 (<title>…</title>) Title Tag

```
<html>
<head>
<title>This is my page</title>
</head>
<body>
</body>
</html>
```

Refresh the page in the browser and see what changes displayed there.

Now copy the file 'yourname.htm' as yourname_2.htm and change the title as "This is my Second page".

Now add the following line between body tags in yourname.htm file.

Example 9.3 (<a> …) Anchor Tag

```
<html>
<head>

<title>This is my page</title>
<a href="yourname_2.htm">My second page</a>

</head>
<body>
</body>
</html>
```

An anchor (<a>…) element is called an anchor because it is used to anchor a URL to some text on a web page.

Now add the following (bold) line in yourname.htm file.

Example 9.4 (
) (A forced line-break)

```
<html>
<head>
<title>This is my page</title>
</head>
<body>
<a href="yourname_2.htm">My second page</a>
<br>
</body>
</html>
```

Now add the following (bold) line in yourname.htm file.

Example 9.5 (<table>…</table>) Table Tag

```
<html>
<head>
<title>This is my page</title>
</head>
<body>
<a href="yourname_2.htm">My second page</a>
<br>
<table border="1">
<tr>
<th>First Column Heading</th>
<th>Second Column Heading</th>
</tr>
<tr>
<td>First Column </td>
<td>Second Column </td>
</tr>
</table>

</body>
</html>
```

`<tr>...</tr>`

Contains a row of cells in a table.

`<th>...</th>`

A table header cell; contents are conventionally displayed bold and centered.

`<td>...</td>`

A table data cell.

Now add the following (bold) line in yourname.htm file.

Example 9.6 (<img>) Image Tag

```
<html>
<head>
<title>This is my page</title>
</head>
<body>
<a href="yourname_2.htm">My second page</a>
<br>
<table border="1">
<tr>
<th>First Column Heading</th>
<th>Second Column Heading</th>
</tr>
<tr>
<td>First Column </td>
<td>Second Column </td>
</tr>
</table>
```

```
<br><br>
<img src="../images/tulips.jpg" alt="tulips">

</body>
</html>
```

<img> is used to insert an image in the document. The **src** attribute specifies the image URL. The required **alt** attribute provides alternative text in case the image cannot be displayed.

9.1 HTML Forms

HTML forms are used to pass data to a server.

A form can contain input elements like text fields, checkboxes, radio-buttons, submit buttons and more. A form can also contain select lists and textarea elements.

The <form> tag is used to create an HTML form:

```
<form  action="url">
.
input elements
.
</form>
```

9.2 HTML Forms - The Input Element

The most important form element is the input element.

The input element is used to select user information.

An input element can vary in many ways, depending on the type attribute. An input element can be of type text field, checkbox, password, radio button, submit button, and more.

The most used input types are described below.

9.2.1 Text Fields

<input type="text" /> defines a one-line input field that a user can enter text into:

```
<form>
First name: <input type="text" name="firstname" /><br />
Last name: <input type="text" name="lastname" />
</form>
```

How the HTML code above looks in a browser:

First name:

Last name:

Note: The form itself is not visible. Also note that the default width of a text field is 20 characters.

9.2.2 Password Field

<input type="password" /> defines a password field:

```
<form>
Password: <input type="password" name="pwd" />
</form>
```

How the HTML code above looks in a browser:

Password:

Note: The characters in a password field are masked (shown as asterisks or circles).

9.2.3 Radio Buttons

<input type="radio" /> defines a radio button. Radio buttons let a user select ONLY ONE one of a limited number of choices:

```
<form>
<input type="radio" name="sex" value="male" /> Male<br />
<input type="radio" name="sex" value="female" /> Female
</form>
```

How the HTML code above looks in a browser:

Male

Female

9.2.4 Checkboxes

<input type="checkbox" /> defines a checkbox. Checkboxes let a user select ONE or MORE options of a limited number of choices.

```
<form>
<input type="checkbox" name="vehicle" value="Bike" /> I have a bike<br />
<input type="checkbox" name="vehicle" value="Car" /> I have a car
</form>
```

How the HTML code above looks in a browser:

I have a bike

I have a car

9.2.5 Submit Button

<input type="submit" /> defines a submit button.

A submit button is used to send form data to a server. The data is sent to the page specified in the form's action attribute. The file defined in the action attribute usually does something with the received input:

```
<form name="input" action="form_action.php" method="get">
Username: <input type="text" name="user" />
<input type="submit" value="Submit" />
</form>
```

How the HTML code above looks in a browser:

Username:

If you type some characters in the text field above, and click the "Submit" button, the browser will send your input to a page called "form_action.php". The page will show you the received input.

9.3 HTML Forms - The Select Element

The <select> tag is used to create a select list (drop-down list).

The <option> tags inside the select element define the available options in the list.

```
<select>
 <option value="toyota">Toyota</option>
 <option value="mercedes">Mercedes</option>
 </select>
```

9.4 HTML Forms - The textarea Element

The <textarea> tag defines a multi-line text input control.

A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier).

The size of a textarea can be specified by the cols and rows attributes, or even better; through CSS' height and width properties.

```
<textarea rows="2" cols="20">
 This is my first HTML program.
 </textarea>
```

9.5 HTML Forms – The Form tag Attributes

Following lines elaborate the <form> tag attributes.

9.5.1 Required Attributes

Attribute	**Value**	**Description**
action	*URL*	Specifies where to send the form-data when a form is submitted

9.5.2 Optional Attributes

Attribute	**Value**	**Description**
accept	*MIME_type*	Specifies the types of files that can be submitted through a file upload
accept-charset	*charset*	Specifies the character-sets the server can handle for form-data
enctype	application/x-www-form-urlencoded multipart/form-data text/plain	Specifies how form-data should be encoded before sending it to a server
method	get post	Specifies how to send form-data
name	*name*	Specifies the name for a form
target	_blank _self _parent _top *framename*	Specifies where to open the action URL

10 Resources

10.1 websites

http://www.w3schools.com/

10.2 Books

Professional PHP Programming by Wrox word press Ltd. **ISBN:** 81-7366-201-0